Article

Games and Culture
2021, Vol. 16(5) 499–518

Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/1555412020914722
journals.sagepub.com/home/gac
SAGE

Making Sense of Abstract Board Games: Toward a Cross-Ludic Theory

Sandra Danilovic[1] and Alex de Voogt[2]

Abstract

The frequent absence of culturally specific, figurative, or decorative markings in abstract board games has challenged theorizations that assume a meaningful representation in the study of games. In accepting this challenge, this article theorizes the historical phenomenon of abstract board games whose nonrepresentational board design and formal rules have transmitted with little change over millennia and across vast expanse. A theoretical framework is outlined for understanding abstract board games—a modular ontology of abstract board games and a typology of player meaning-making in abstract board games. It is argued that the reproducibility and transferability of abstract board games as self-sufficient and reliable formal systems that players share independently from culturally specific meanings and materials may contribute to their dispersal. It is in this interaction between the cross-cultural/reliable and local/variable semantic structures of abstract board games that game studies from a historical or archaeological perspective may meet literary and social science perspectives.

Keywords

abstract board games, formal rules, meaning-making, archaeology, ontology

In the study of games and play, a specific set of board games that has been played for 5 millennia and that is especially dominant outside the (post)industrialized world has

[1] Wilfrid Laurier University, Brantford, Ontario, Canada
[2] Drew University, Madison, NJ, USA

Corresponding Author:
Sandra Danilovic, Wilfrid Laurier University, 73 George Street, Brantford, Ontario, Canada N3T 2Y3.
Email: sdanilovic@wlu.ca

been excluded from theoretical consideration. Studies of literary theory, sociocultural anthropology, and other social sciences have largely ignored abstract board games as they are attested in antiquity and, nowadays, in most nonindustrialized societies. These games, such as checkers, Egyptian *senet*, and mancala games, consist of arrangements of squares or holes often with little to no additional markings or culture-specific elements. In their basic layout, these games do not refer to the cultural or local context in which they are played. As a result, they spread across time and space, unaffected by sociopolitical borders, and have remained largely unaltered in their appearance and in their mode of play (Crist, de Voogt, & Dunn-Vaturi, 2016; de Voogt et al., 2013). They have been appropriated by many playing groups without a conscious understanding of their history (e.g., de Voogt, 2018).

The absence or optional presence of culturally specific, figurative, or decorative markings in abstract board games has challenged the application of theoretical frameworks that assume a meaningful representation and that situate games and play accordingly. For example, game scholars have argued that games cannot be separated from the player's interpretations, termed "meaningful play" (Salen & Zimmerman, 2004). One assumption built into this concept of meaningful play hinges on interaction that presumes a reciprocal influence—that players are not only affected by the games they play but that they also affect the games themselves. If players do not change abstract board games through cultural play practices, how do we theorize the historical phenomenon of abstract board games whose nonrepresentational board design and formal rules have transmitted without significant change over thousands of years and across vast expanse?

We argue that the assumption about games' *variable qualities*, which are indeed determined by players, holds for abstract games but that this is not the case for their *reliable properties*[1]—their self-sufficient formal systems, which generally remain unaffected by players and play environments. The modularity of formal systems—their reproducibility and transferability as mathematical structures and largely unambiguous rule sets that players share independently of culturally specific meanings—may contribute to the cross-cultural and cross-temporal dispersal of abstract board games. In other words, while abstract board games may be "placeholders" that players replace with their own culturally relevant meanings and materials, they are neither modified by local play environments nor impacted by the player's imaginative import; rather, they travel without the cultural, material, or individual "baggage" of play spaces. Abstract board games are thus distinct from nonabstract games because they harness culturally inclusive play practices—players and play communities imbue these games with their own set of unique meanings and cultural values. As a result, their reliable formal systems, flexible visual properties, and nonessential materials comprise what we call, the *distinct modular ontology of abstract board games*.

This article contributes to a greater understanding of abstract board games. First, our modular ontology of abstract board games sheds light on their unique features that may explain their cultural transmission across time and space. Second, our typology of intrinsic and extrinsic game semantics provides a better understanding of the types of player meaning-making in abstract board games—cross-cultural/reliable and local/variable. Third, this typology highlights important distinctions between the player's logico-analytical and imaginative interpretations of the game, which can be productive for studying their overall interactions in a variety of abstract game types and genres.

We draw on empirical studies in cultural anthropology, game studies, and the philosophy of cognitive science to theorize abstract board games. We first identify the role of abstract, "traditional," and nonproprietary games in the broader social sciences, humanities, and game studies literatures. We describe one case study, checkers in the Marshall Islands, which illustrates how the rule set of checkers is not specifically meaningful for the cultural context in which this game is played. Next, we discuss representational and formalist theories of rules in relation to meaningful play and sketch out the distinct modular ontology of abstract board games. This ontology may be useful for building a more substantive theory of abstract board games in the future.

The Category of Abstract Board Games

In cultural anthropology, the main theoretical idea that has extended to abstract games is that by Roberts et al. (1959) who hypothesized that "simple societies" should not play strategic games—a Eurocentric claim that was rather broadly defined and, therefore, includes most if not all board games—and should resist borrowing them. This work was later extended by Chick (1998) but criticized especially by those studying board games (see Crist, 2019; de Voogt, 2017b; Townshend, 1980) to the extent that the main hypothesis is no longer tenable (Chick, 2017; de Voogt, 2017a). Instead, theories developed in archaeology have been successfully applied to abstract board games addressing questions of cultural transmission (de Voogt et al., 2013; Hall & Forsyth, 2011), liminality (Crist, de Voogt, & Dunn-Vaturi, 2016), and social complexity (Crist, 2016, 2019; Rogersdotter, 2011).

The category of abstract board games includes abstract strategy, "traditional," and nonproprietary games that have been played over thousands of years. Abstract strategy games are a category of games that has been used to define those strategy games that do not rely on a theme and in which a theme is not important to the player experience (Thompson, 2000). A group of games inventors have specialized in abstract strategic games leading, for instance, to books of chess variants (e.g., Pritchard, 1994).

Of all these games, chess has received extensive interest, also from the social sciences, due to the figurative pieces that hint at a theme of kingdoms and warfare. This theme is also apparent in its many variants found around the world (Murray, 1913) but arguably social scientists still classify the game as an abstract strategy game. Thompson (2000) claims that chess is not defined by its figurative content of kingdoms and war: "apart from the names of the pieces there is nothing about the game itself suggesting war; it is more suggestive of geometric patterns" (para. 5). Rather, chess, similar to other abstract games and self-contained puzzles, does not rely on the player's storytelling but is "an exercise in logical thought," states Thompson. The logical and strategic prowess of chess-playing artificial intelligence computers such as Deep Blue illustrates this emphasis.

While the abstract strategy games became a category for games inventors, the games discussed in this article exclude invented and proprietary games. Games with an educational or moralistic aim started to emerge in the Renaissance and more widely in the 19th century. They were sold with rule sets rather than being home-made with rules already understood by the players. Later inventions became part of an industry that expanded the thematic development of games that was mostly geared toward families and children (see Hofer, 2003; Parlett, 2018; Whitehill, 1999). The opposite of invented and proprietary games has often been termed "traditional." The latter term is problematic in multiple ways but it suffices to point out that even proprietary games may fall into this category especially as they become appropriated by different societies around the world (e.g., de Voogt, 2019).

Within these extant definitions, there is a group of abstract games, all board games, that can be defined historically: They are all abstract board games, with the possible exception of chess, and were played prior to the Renaissance. This definition excludes card games and board games of later times that could benefit equally from inclusion in theoretical approaches to games. However, these latter games are considered a subcategory of the abstract games discussed here.

The set of abstract board games relevant to this article that was played prior to the Renaissance consists of configurations of playing squares or holes and sets of playing pieces that usually have one or two categories only. This is the main reason to regard chess as different from all these games as it has, for instance in the Western variant, already six different figurative pieces and two colors, a diversity not found in any of the other games. The names of pre-Renaissance board games in the literature that describes them already point to the abstract nature of their appearance as they include, for instance, the game of 20 squares, the game of 58 holes, the game of 5 lines, the game of 30 squares, three-men's morris, and nine-men's morris (see Figure 1). Mancala games are a group of games for which all playing pieces are identical as different sides share the same playing counters. The word "mancala" is thought to derive from "to move" (naqal) in Arabic. Mancala is also called a "rational game" (*La'b hakimi*) and "intelligent game" (*La'b akila*) as played in Syria (Culin, 1894, para. 4). As Culin states, "success in (mancala) depends largely upon the skill of the players" (para. 4). These board games are also found scratched in rock surfaces that prevent any figuratively meaningful interpretation of the board design other than the identification of the game itself (see, e.g., Charpentier et al., 2014; Rogersdotter, 2015; see Figure 2); and although sculptured mancala game boards have had an important function in gift exchange (Walker, 1990), they are less frequently included in play.

Figure 1. Siga (also Seeja or Dhala), with a game board commonly consisting of five rows of five playing fields, Sai Island, Sudan (photo courtesy of Alex de Voogt, 2011).

Figure 2. Mancala game consisting of two rows of seven cups, Palmyra, Syria (photo courtesy of Alex de Voogt, 2009).

In antiquity, three-dimensional examples of boards are mostly found in elite tombs that have helped to identify the associated gaming materials (e.g., Tait, 1982) while most other three-dimensional game boards still remain undecorated and without any cultural clues other than the context in which they were found (see, e.g., Hall, 2016; Hillbom, 2011; Swiny, 1980). These board games are attested in Ancient Egypt and Mesopotamia as well as in Roman, Mayan, Ottoman, and other ancient and Medieval contexts (see Crist, Dunn-Vaturi, & de Voogt, 2016; Schädler, 2007; Voorhies, 2017). The board games dating to Medieval times are still played in several places around the world and have been for extended periods of time. They are a fixture of play in many societies in addition to sports and later proprietary games using a host of different materials and ways of play.

The abstract board games in question are predominantly played by two adult players at a time. They mostly do not include children and almost always feature one or more fixed sets of playing rules and board configurations. The process of negotiating rules and playing space is largely absent, limiting the possibilities of research for those interested in game formation. In games and gaming research, however, the presence of fixed rules is not necessarily a limiting aspect. On the contrary, rules and their development and understanding is part of theoretical discourse in a way that may apply to abstract board games as well. In other words, the parts that abstract board games share with games in general are an obvious starting point for exploring theories that cross the research divide currently observed between games from the pre- and post-Renaissance periods. The exclusive problem of abstract board games lies in the absence of a theme or any other figurative or culturally specific interpretation.

Checkers in the Marshall Islands: A Case Study

Consider first how a particular case, checkers in the Marshall Islands, illustrates the paradox of meaningful play—how the players' interpretations and the context in which this game is played do not impact the formal rules of the game. This game's distinct rule set does not reveal anything meaningful about the cultural context in which this game is played.

The Marshall Islands is a nation located in Micronesia in the western Pacific Ocean. The islands feature an active community of competitive players of checkers, known as *jekab* in Marshallese, who use a rule set that is particular to the region. The game and its players were part of a study conducted in 2017/2018 that included a detailed description of the playing rules as well as the cultural significance of the game in Marshallese society.

The rules specific to the Marshall Islands are also attested in neighboring Tuvalu, Nauru, Kiribati, and in the culturally unconnected Trobriand Islands (Sergio Jarillo de la Torre, personal communication, May 19, 2017) located in Melanesia, which is located further south. The distinction between *jekab* and checkers outside of this region hinges on the rules for promoting a piece to a king, which is a significant element in the proceedings of the game. When a checker piece reaches the far end of the board, it becomes a king, but if there are more captures to be made, the crowning of the king either has to wait or the piece will have to rest on the far side before it can make captures as a king in a subsequent move. In Marshallese checkers, however, the piece transforms to a king mid-move and will continue as a “long” king as soon as it has passed the far row. There are particularly few regions where the same rule set can be found; they are attested mainly in the abovementioned western Pacific and Russia (Ratrout, 2019).

There is no historical connection with Russia, and even within the Pacific, there is no known contact between the Trobriand Islands and its Micronesian neighbors. Historically, this set of rules is likely to have been common among American soldiers even though we have no records of their playing rules to confirm this. American Pool checkers is identical to Marshallese checkers with the exception of this one rule but Pool checkers did not become standardized until the 1960s, while other variations may have been played in the early 1940s and spread during the American military presence in the Second World War.

Attributing meaning to this rule set has additional complications. The Marshallese are unaware of other variations of the game and the introduction to the islands of this game of checkers is not remembered even by the senior players. Instead, they maintain the game has been on the islands since time immemorial. The rules for making a king are not seen as unusual or specific to the Marshall Islands, it is simply how the game is played.

The game of checkers has gained prominence in Marshallese society but not because of its seemingly unusual rule set. It is featured in yearly tournaments held during cultural celebrations on multiple islands in the archipelago, while the activity is considered an integral part of public life as it is witnessed in public places on the islands. Together with canoe peddling, basket weaving, coconut husking, and fire making, checkers play features in a yearly cultural festival that dates back at least a decade. In 2016, for instance, there were four checkerboards being used for a knockout tournament with cash prizes for the winners in such a festival. No other board or card game features in these festivals and checkers is officially considered part of Marshallese cultural heritage.

The role of checkers as a social lubricant (Crist et al., 2016) can be illustrated with the diversity of players that gather daily at the checkers board in central Majuro, the capital on the main island. One afternoon, there were seven players, 23 years apart, from seven different atolls: Kwajalein, Rongelap, Ailinglaplap, Ebon, Maloelap, Ujae, and Arno. They all learned their skills outside of Majuro, depending on where they lived in their early 20s. They said that work or school had brought them to Majuro where they now interacted with fellow players from around the country. Their professions range from schoolteacher and museum director to fireman, policeman, small business owner, and gardener. Interviews confirmed that the game’s social function across geographical, social, and generational divides was acknowledged and valued by the participants. At the same time, this social interaction was limited to the playing board as players rarely socialized or interacted outside the context of competitive play.

In short, playing checkers in Majuro is a meaningful activity that is not informed by the unique set of playing rules. While the rules are of historical significance and speak to the cultural transmission of the game of checkers, they do not inform us about the meaning of this cultural practice in the Marshall Islands or its neighbors. We need to turn to interdisciplinary perspectives on rules to gain insight into why this might be the case.

Interdisciplinary Perspectives on Rules

Two different perspectives are presented here that are related to rules and that may contribute to a more universal perspective for theorizing abstract games. The first features rules as meaningful representations and relates to an exchange between the formal system of the game and the outside world. This perspective is supplemented by studies that consider rules as abstract architecture. The examples help us recognize the fundamental distinction between the player's interpretations of the game and the game's formal system, the latter of which is not impacted by players and communities.

Rules as Meaningful Representations

A rule is a condition that the player must meet resulting in an outcome; violating this rule produces a different outcome. Rules limit player action (Salen & Zimmerman, 2004, p. 122) and "describe what players can and cannot do" (Juul, 2011, p. 55). Several theories of rules in game and play studies interpret rules as representational, figurative, or thematic in nature (i.e., Bogost, 2007; Juul, 2011). The assumption in these theories is that rules are almost never divorced from the player's imaginative interpretations and that the player is always making sense of game rules according to what they symbolically represent in the world.

The idea that play is meaningful can be traced to Huizinga (1949/2002). "All play means something," Huizinga states (p. 1). Salen and Zimmerman's (2004) concept of "meaningful play" extends Huizinga's insights: "meaningful play emerges from the interaction between players and the system of the game, as well as from the context in which the game is played" (p. 33). This context might be the sociocultural realm in which the game is played or the community of players that give meaning to social play practices, such as the example of checkers in the Marshall Islands.

Recently, video game rules have set the tone in the game studies literature for theorizing the relationship between rules and meaning. Juul (2011) frames video games as ruled fabulized worlds: "to play a video game is therefore to interact with real rules while imagining a fictional world" (p. 1). Juul argues that gameplay is simultaneously real and fictional or "half-real." Bogost (2007) argues for the expressive power of video games as a form of "procedural rhetoric" that persuades through "rule-based representations and interactions" (ix). For Bogost, rules in video games help the player interpret the game world as a specific, material, and embodied realm of human experience (p. 241).

Given the above perspectives, the player is always meaningfully interpreting game rules and drawing comparisons between the game and the outside world. We could argue that even players of abstract board games assume some representational or thematic content when they call their game pieces "soldiers," "men," "dogs," or "seeds" while they use verbs such as "capture," "kill," and "sow" in their playing vocabulary. Murray (1952) argued that most historical board games represent human activity and classified them as war games, hunt games, and so forth. Flanagan and Nissenbaum (2014) suggest that playing one of the abstract mancala board games is similar to "sowing seeds in a field" (p. 18) which invokes agrarian societies. However, mancala is also played, for instance, in fishing communities with boards in the shape of fish and cowrie shells as counters (de Voogt, 1997). One could argue that players are free to add a thematic layer of their choice as abstract board games rarely define the appropriate theme a priori. Indeed, the same abstract game may conjure up different names for pieces and moves.[2] The storied world seems optional and possibly of different kinds, setting abstract games apart from their more guided siblings.

One problem with some of these representational theories of rules is their conflation of meaning with the game's formal system, which do not help the historical phenomenon of abstract board games. These theories confuse the reliable properties of abstract board games (their intrinsic semantics) with their variable qualities (their extrinsic semantics). Flanagan and Nissenbaum (2014) point to this conflation by suggesting that extrinsic meaning is not embedded in the formal rules of the game: "(these values) are not enshrined in the game's rules"; rather, they are "embedded in the materials used in the boards, the presentation of the game, and the context created by the community where it is played" (p. 18).

The historical crossing of borders by abstract games is useful to criticize extant studies that insist on this conflation of formal rules and meaning. For instance, Townshend's (1986) attempt to interpret the rules of bao, as it is found on the island of Lamu off the coast of Kenya, as socioculturally meaningful contradicts the spread of the game along the Swahili trade routes. Concurrently, the epistemic separation of formal rules and meaning supports the idea that games such as the Egyptian game of *senet*, gained extensive, in this case, religious significance while crossing contexts of play within Egypt; at the same time, it continued to exhibit casual play on graffiti boards; its significance crossed time as *senet* gained religious significance mostly in later periods, that is, the New Kingdom; and it crossed regions as the game spread to Cyprus *without* this religious significance (Crist et al., 2016, pp. 41–59).

The historical evidence of *senet* points to the spread of the formal systems of abstract board games that occurs separately from the dispersal of the game's extrinsic or contextually defined features. Differentiating between the game's intrinsic properties such as its formal rules and the player's extrinsic meaning-making is productive for teasing out the reliable properties of abstract board games from their variable qualities, the former of which facilitate their cross-cultural dispersal and remain unaffected by players and contexts.

This is a counterintuitive epistemic move. Salen and Zimmerman (2004) warn that “the internal logic of the game is not something that can be completely severed from the ways that the game exchanges information with the outside world” (p. 86–87). We claim that this exchange between the formal system of the game and the outside world—“meaningful play”—is not fully reciprocal, which is why we need to consider *how* and *why* abstract board games might be self-sufficient from the cultural spaces in which they are played.

To examine the independence of abstract board games from their culturally holistic occurrence, we turn to formalist perspectives of rules, notably those of philosopher of cognitive science and artificial intelligence, John Haugeland.

Rules as Abstract Architectures

Abstract architectures of rules have been formulated in different ways that all speak of systems. Starting with the idea of formal systems, abstract games could be understood as modular systems or virtual machines. These perspectives confirm that uncritically conflating the formal system with culturally specific interpretations creates confusion rather than clarity.

Games as formal systems. Formalist perspectives examine games as self-contained formal systems prior to their interaction with the world. All games are formal systems (e.g., Crawford, 1984; Fullerton, 2019; Salen & Zimmerman, 2004). Haugeland (1981) emphasizes the intimate connection between games, mathematics, and logic: “considered only *as formal systems* [emphasis in original], games, logic, and mathematics are all equally meaningless, and entirely on a par” (p. 10). Haugeland (1985) affirms that “many familiar games—among them chess, checkers, Chinese checkers, go, and tic-tac-toe—simply are formal systems” (p. 48).

Abstract board games as modular systems. As formal systems, abstract board games are self-contained and as such, unaffected by the “messiness” of the concrete environment in which they are played: “it makes no difference to a chess game, as such, if the chess set is stolen property or if the building is on fire or if the fate of nations hangs on the outcome—the same moves are legal in the same positions, period” (Haugeland, 1985, p. 50). Chess’s self-sufficiency ensures that its formal rules do not depend on the game’s external, contextual, and environmental influences, which includes the player’s sense-making: “since formal systems as such are self-contained, nothing about them can depend on meanings” (Haugeland, 1985, p. 73). In other words, “meaning is not a formal property—because (roughly) meanings relate to the ‘outside world,’” says Haugeland (p. 50). Caillois (2006) also suggests that precision, not meaning, is the objective of rule-construction: "man merely adds refinement and precision by devising rules" (p. 132).

The self-sufficiency of formal systems ensures their modularity—allowing the system to be shared independently of the player's interpretations and play contexts. Indeed, formal systems can be implemented and rendered in a variety of material supra-structures. Frasca (1999) invokes this modular quality of formal systems in his definition of ludus: "ludus have a defined set of rules. These rules can be transcribed, and easily transmitted among different players."

This "medium independence" (Haugeland, 1985) ensures that the thematic value of the tokens and board are nonessential to the formal identity of the abstract board game, which invokes the role of variables in computer programming—the computer program's ability to summon the user's/player's distinct inputs (values). Haugeland uses the following example to emphasize this modularity: Texas millionaires playing "helicopter chess" from their opposing penthouses, "using thirty-two radio-controlled helicopters and sixty-four local rooftops" (p. 58). Despite the use of helicopters as tokens and rooftops as a chessboard, they are still playing a game of chess.

Abstract board games as virtual machines. When formal systems interact with the real-world, they become meaningful *to* players and communities. According to Haugeland (1985), "to interpret a system of marks or tokens as symbolic is to make sense of them all by specifying in a regular manner what they mean" (p. 94).

An important insight that Haugeland supplies with respect to the generation of meaning is the idea that rule-following ensures an interpretation that is intelligible and that "makes sense" to humans: "simply playing by the rules is itself a surefire way to make sense" (p. 106). Formal systems and their tokens, which are coherent structures, are thus meaningful because of their order. Humans make sense of that which has some kind of orderly and nonrandom structure. "Arithmetic salad does not make sense . . . ; and that, primarily, is why random interpretation schemes don't really give *interpretations* [emphasis in original] at all" (Haugeland, 1981, p. 27). Formal systems, and by extension, abstract board games, are meaningless on their own, prior to referring to the world, or being *about* it. Meaning "takes form," is organized and thus "makes sense" to players via the coherent (logical) structure of the game.

Abstract relationships posited in the formal rules circumscribe the essences of interaction, which are modular and reproducible, whereas the variable player inputs circumscribe the specifics of interaction—the latter of which allow diverse meanings to emerge. Haugeland (1981) describes this formal property of rules: "the rules are still truth-preserving, no matter what specific interpretation is given to the variables" (p. 24). In other words, the game's formal system does not itself furnish the specificity of the world, the players interacting with its coherent structure do. Abstract board games are in that sense culturally customizable and adaptable, that is, flexible to take on any culturally circumscribed meanings. Abstract board games determine meaning a posteriori. As such, they are virtual machines that enable culturally specific player interpretations to dynamically emerge from their abstract architectures.

A game of chess illustrates this idea of virtuality, as per Haugeland's helicopter chess example, through the various cultural meanings it has acquired contingent on context and players. Traditional chess is considered to be a game of war (Flanagan & Nissenbaum, 2014, p. 42; Murray, 1913, p. 25). Nonetheless, its rich history speaks to the sheer variability of meanings imposed on its board, tokens, and rules. Chess originated in India prior to its transmission to the Arabic world and Europe (Adams, 2006, p. 2; Murray, 1913, p. 51). One noteworthy aspect about chess' cultural transmission is that the chess queen did not exist prior to European medieval culture. In India, Persia, and the Arabic world, "all the human figures were male" (Yalom, 2005, p. xviii). Not only was the queen a male counselor, but the bishop was an elephant, and the horse was a knight or horseman (see Adams, 2006, p. 3; Murray, 1913). "Almost all European cultures changed the gender of the piece in what can only be a conscious attempt to map its own social structure onto the game" (Adams, 2006, p. 3). Even though the queen has rarely been more powerful than a king in the monarchies of the West, "(the king) is the crucial figure, even if (the queen) is more potent" (Yalom, 2005, p. xviii).

According to medieval chess scholar Jenny Adams, European variations on chess, as reflected in literature, essays, and rule books "reflect different understandings of the game" (p. 8). One (rather common) interpretation is chess as a political fantasy of medieval social hierarchy (p. 7) while other interpretations on the themes of war, kingdoms, and conquest include chess as a model of citizenship and even a code of sexual desire (p. 8). The tokens have also been infused with different power values; as Adams points out, even the pawns have been heralded as tricksters with powers of deception that could bring down a "great man" (p. 162), presumably a king. Adams also underscores the moralistic objectives assigned to chess, as per Benjamin Franklin's 1779 essay "The Morals of Chess," which reconceives chess as a self-improvement tool—a departure from the chess-as-political-fantasy trope (p. 164). The variability in meaning is in part determined by the player's selective and preferential (heuristic) reading which may retain some culturally symbolic elements while excluding others. Adams confirms:

> A similar detachment allows contemporary Western cultures to use chess, a game still populated by kings, queens, and knights, to represent a variety of things from terrorist attacks to "sweeps week" on television, from the canniness of "historical materialism" to, as Ferdinand de Saussure uses it, the ways language systems work (p. 164).

It thus seems that the kingdom theme is embedded in the formal rules of the game. What if we replace a human kingdom with an "animal kingdom" in chess? All that is embedded in the formal rules of chess is a hierarchy, not a monarchy. But, we might counterargue, chess is not an ahistorical game—it is historically circumscribed to connote human kingdoms. This historicity is indeed a part of chess as a cultural practice, with one caveat: The kingdom theme is *optional* and can be replaced by a different symbolic system, as Adams and Yalom have observed. The hierarchy of chess tokens is woven into the structure of chess according action values to them, but the successful play of a game of chess does not hinge on the kingdom theme. To further illustrate this, the Bauhaus designer Josef Hartwig designed a chess set in 1924 with chess pieces that are a visual shorthand for the game's legal moves (horizontal, vertical, diagonal and one, two, or more squares depending on the chess piece) instead of the usual figurative pieces. Through his procedural redesign of chess, Hartwig amplifies the primacy of logical reasoning and strategic play over the player's imaginative import of kingdoms, conquest, and war in successfully playing a chess game (Hartwig, (n.d.), Museum of Modern Art, New York). Indeed, as Adams (2006) confirms, chess is an "ultimately flexible" game, "that can be used in multiple symbolic systems yet perhaps because of this malleability, now both full of meaning and simultaneously meaningless" (p. 164).

Salen and Zimmerman (2004) expand on the relationship between formal rules and meaning. They categorize game systems as three overlapping types: formal rule systems, experiential systems, and contextual (culturally embedded) systems, which together constitute how a player experiences the game. Cognitive scientist Douglas Hofstadter (1979/1999) similarly highlights the permeable boundaries of formal systems: "life is composed of so many interlocking and interwoven and often inconsistent 'systems'" (p. 38). Hofstadter, however, claims that "it is very important when studying formal systems to distinguish working within the system from making statements or observations about the system" (p. 38). Hofstadter is making a distinction between a game's intrinsic semantics—the game's formal system; and the game's extrinsic semantics—the contextual and experiential system, where the player's imaginative import plays a role in both fictional game worlds and abstract board games.

The above theoretical and practical examples help us recognize the fundamental distinction between the player's diverse interpretations of the game and the reliable aspects of the game's formal system, which are generally not impacted by players and communities. To that end, and especially in the case of abstract board games, uncritically conflating the formal system with culturally specific interpretations creates confusion instead of clarity.[3]

A theoretical framework for understanding abstract board games

After the above exploration of modular formal systems, we outline a theoretical framework for understanding abstract board games comprising of two components—a *modular ontology of abstract board games* and a *typology of player meaning-making in abstract board games*—which both describe the cross-cultural/reliable and local/variable semantic structures of abstract board games. The

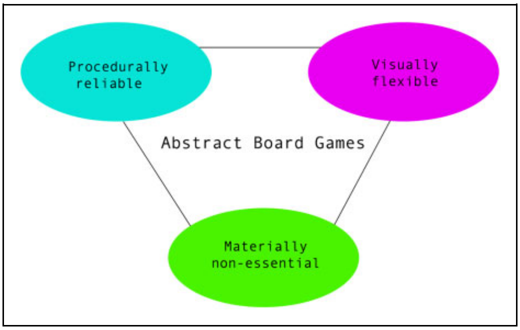

Figure 3. The modular ontology of abstract board games.

importance of this framework rests in its potential to bridge multiple disciplinary perspectives on abstract board games and contribute to ongoing conversations in game studies.

A modular Ontology of Abstract Board Games

It is now possible to formulate a modular ontology of abstract board games that may also explain their diffusion across time and space (see Figure 3). This ontology is comprised of three context-independent properties of abstract board games: their *procedural reliability,* their *visual flexibility,* and their *material nonessentiality.* The modular ontology of abstract board games maps three activities that abstract board game players engage in: universal cognitive processes, culturally specific interpretations, and local material practices used to recreate the board and tokens.

Abstract board games are *procedurally reliable*. Irrespective of cultural context, players can transcribe and reproduce (map) their formal rules as they were originally construed and follow them. A player can map any material object to the proper values of tokens according to the game's interpretation scheme, as was exemplified by the Josef Hartwig chess design. As such, the abstract visual properties make these games *visually flexible*—players reimagine the board design and tokens from their own cultural standpoint. Either way, the legal moves are still the same. Whatever historically circumscribed "world-aboutness" players attach to these tokens and the board, this aboutness has no effect on the formal rules of the game.

Finally, abstract board games are *materially nonessential.* The material composition of the board and tokens are inconsequential to the formal rules and do not change the game's intrinsic formal identity. Players choose materials that are local to their environment and/or have symbolic value to their community, but these choices do not impact the game's formal identity. For example, in mancala games, a fishing community may use shells as tokens whereas a farming community may use seeds (de Voogt, 1997).

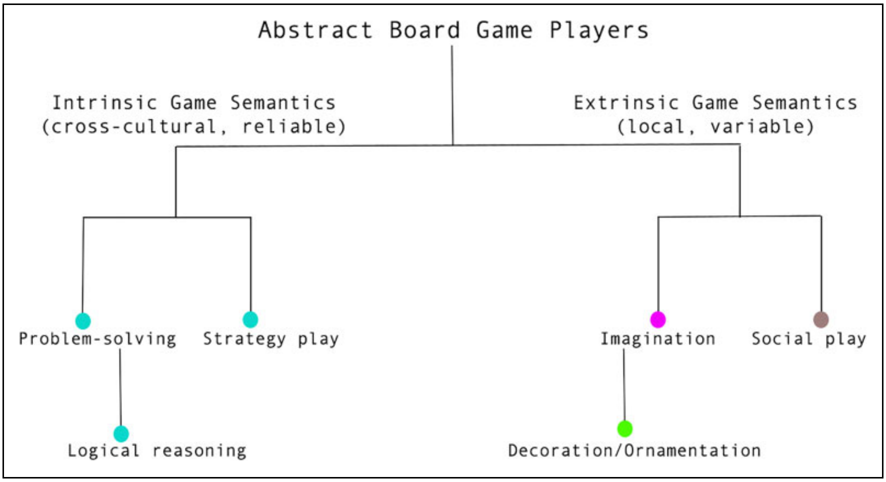

Figure 4. The typology of player meaning-making in abstract board games.

A typology of Player Meaning-Making in Abstract Board Games

The *intrinsic game semantics* or the player's logico-analytical interpretations of the formal rules may cross borders in the sense that all players use cognitive processes to make sense of the game rules. The *intrinsic game semantics* account for the structure of the player's understanding of the formal rules—the player's intellectual import such as logical reasoning and problem-solving.

The *extrinsic game semantics* are the player's symbolic interpretations of the game's formal rules, board design, tokens, and material properties. These semantics are the culturally specific and variable (local) interpretations that individuals and communities bring to gameplay. They are culture-, community-, and individual-specific and may differ from one player/community to another player/community (see Figure 4).

Distinguishing between the game's intrinsic and extrinsic semantics helps game scholars get a better sense of how meaningful play emerges in abstract board games. A way to relate to this difference from a game design perspective is by considering the common disconnect between the game designer's intentions when they are designing a game and the player experience, the latter of which does not often align with those intentions, precisely because players bring their own diverse interpretations to games that designers could not have predicted. This form of semantic disconnect is bridged through playtesting and user experience design.

If we epistemically distinguish between the abstract board game's formal system and the player's imaginative import, it is possible to better integrate abstract board games research in the theoretical discourse of play and modern gaming. Conflating formal rules and meaning-making in abstract board games is theoretically problematic. It presumes the formal system is less likely to cross temporal and sociocultural borders without change.

Conclusion

Understanding the extrinsic game semantics of abstract board games—the player's imaginative import—is an especially productive exercise if the formal rule system is generalized across contexts in which the game is found. This latter element is counterintuitive in play research, such as that of video games, where the study of the game structure itself already provides a meaningful and generalizable starting point for a later contextual analysis. As a result, the study of meaning in abstract board games is confined to a specific players' context, such that only the context and not the game itself is object of study. The abstract board game merely facilitates the interaction and may not even limit this encounter differently from other abstract board games, a question that is now particularly relevant to explore. Although the formal rules are to be distinguished from the extrinsic meaningful activity, each context may reveal different interactions between the two. It is mainly in this interaction between the cross-cultural/reliable and local/variable semantic structures of abstract board games, that the study of abstract board games from a historical and archaeological perspective may meet the studies from a sociocultural, literary, or other social science point of view.

The cross-cultural transmission of abstract board games as a historical phenomenon has possible further implications. Embedded in their respective cultural (local) practices, abstract board games might have been instrumental in the dissemination of systems of thought. They may have contributed to the cross-cultural dispersal of logic and mathematics among Asia, the Middle East, Europe, and the Americas and influenced modern systems of thought. For instance, Gobet et al. (2004) argue that the formal systems of board games allowed for experiments that led to generalizable insights in cognitive processes such as decision making, perception, and memory. If systems of thought are enacted through cultural performance (i.e., cultural rituals and customs), then, in their algorithmic beauty, abstract board games may have found a way to reproduce the essences of that performance.

Abstract board games are rigorous formal systems that humans have created as early as 5 millennia ago. Studying their formal properties may provide further insight into the origins of games as computational artifacts of play. It is up to game studies to delineate how they might study abstract board games as a rich cross-cultural play practice that simultaneously transcends and integrates cultural differences.

Acknowledgments

We thank the editor and two reviewers for their constructive feedback and suggestions. We are especially grateful to the government of the Marshall Islands for issuing a research permit as well as facilitating interviews and observations of checkers play. Special thanks go toward to the people of the Alele Museum, specifically director Melvin Majmeto, and the staff of Robert Reimers whose help has been particularly valuable. Finally, we wish to thank Espen Aarseth for facilitating the discussion that served as an inspiration for this article.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Sandra Danilovic https://orcid.org/0000-0002-2453-5689

Notes

1. We are drawing on philosopher of cognitive science and artificial intelligence John Haugeland's (1985) discussion of the "reliability" of digital systems, which points to their *likelihood* to succeed in use—their *tolerance* for variation and error in the real-world. For example, "in basketball, a basket is a basket is a basket; and a miss isn't" (p. 54–55).
2. For the use of "dogs" for playing pieces in antiquity and "pack of dogs" for a game name, see Crist, Dunn-Vaturi, and de Voogt (2016, pp. 63, 67).
3. Philosopher of mind John Searle (1980) gives an example of this type of confusion, such as when humans mistakenly attribute intentionality, a human ability, to nonintentional systems such as robots and artificial intelligence (p. 9).

References

Adams, J. (2006). *Power play: The literature and politics of chess in the late middle ages*. University of Pennsylvania Press.

Bogost, I. (2007). *Persuasive games: The expressive power of videogames*. MIT Press.

Caillois, R. (2006). The definition of play and the classification of games. In K. Salen & E. Zimmerman (Eds.), *The game design reader: A rules of play anthology* (pp. 122–155). MIT Press.

Charpentier, V., de Voogt, A., Crassard, R., Berger, J.-F., Borgi, F., & Al-Mashani, A. (2014). Games on the seashore of Salalah: The discovery of mancala games in Dhofar, Sultanate of Oman. *Arabian Archaeology and Epigraphy*, *25*, 115–120.

Chick, G. (1998). Games in culture revisited: A replication and extension of Roberts, Arth, and Bush (1959). *Cross-Cultural Research*, *32*(2), 185–206.

Chick, G. (2017). Comment on 'strategic games in society: The geography of adult play.' *International Journal of Play*, *6*(3), 319–321.

Crawford, C. (1984). *The art of computer game design: Reflections of a master game designer*. Osborne/McGraw-Hill.

Crist, W. (2016). *Games of thrones: Board games and social complexity in Bronze Age Cyprus* [PhD dissertation, Arizona State University]. https://repository.asu.edu/items/40805

Crist, W. (2019). Playing against complexity: Board games as social strategy in Bronze Age Cyprus. *Journal of Anthropological Archaeology*, *55*. https://doi.org/10.1016/j/jaa2019.101078

Crist, W., de Voogt, A., & Dunn-Vaturi, A.-E. (2016). Facilitating interaction: Board games as social lubricants in the ancient Near East. *Oxford Journal of Archaeology*, *35*(2), 181–198.

Crist, W., Dunn-Vaturi, A.-E., & de Voogt, A. (2016). *Ancient Egyptians at play: Board games across borders*. Bloomsbury Academic Press.

Culin, S. (1894). *Mancala, the national game of Africa* (pp. 597–611).National Museum. https://healthy.uwaterloo.ca/museum/Archives/Culin/Mancla1894/index.html

de Voogt, A. (1997). *Mancala board games*. British Museum Press.

de Voogt, A. (2017a). The absence of games in the presence of claims. Reply to Garry Chick. *International Journal of Play*, *6*(3), 322–323.

de Voogt, A. (2017b). Strategic games in society: The geography of adult play. *International Journal of Play*, *6*(3), 308–318.

de Voogt, A. (2018). Kiribati game development: Cultural transmission, communities of creation, and marketing. *The Journal of Pacific Studies*, *38*(1), 26–41.

de Voogt, A. (2019). Traces of appropriation: Roman board games in Egypt and Sudan. *Revue Archimède: Archéologie et Histoire Ancienne*, *6*, 35–45.

de Voogt, A., Dunn-Vaturi, A.-E., & Eerkens, J. W. (2013). Cultural transmission in the ancient Near East: Twenty squares and fifty-eight holes. *Journal of Archaeological Science*, *40*, 1715–1730.

Flanagan, M., & Nissenbaum, H. F. (2014). *Values at play in digital games*. MIT Press.

Frasca, G. (1999). Ludology meets narratology: Similitude and differences between (video) games and narrative [in Finnish]. *Parnasso*, *3*, 365–71. http://www.ludology.org/articles/ludology.htm

Fullerton, T. (2019). *Game design workshop: A play centric approach to creating innovative games* (4th ed.). CRC Press.

Gobet, F., de Voogt, A., & Retschitzki, J. (2004). *Moves in mind: The psychology of board games*. Psychology Press.

Hall, M.A. (2016). Board games in boat burials: Play in the performance of migration and Viking Age mortuary practice. *European Journal of Archaeology*, *19*(3), 439–455.

Hall, M. A., & Forsyth, K. (2011). Roman rules? The introduction of board games to Britain and Ireland. *Antiquity*, *85*, 1325–1338.

Hartwig, J. (n.d.). *Chess set 1924*. Museum of modern art. https://www.moma.org/collection/works/4240

Haugeland, J. (1981). Semantic engines. In J. Haugeland (Ed.), *Mind Design* (1st ed., pp. 1–34). MIT Press.

Haugeland, J. (1985). *Artificial intelligence: The very idea*. MIT Press.

Hillbom, N. (2011). *Minoan games and game boards: An archaeological investigation of game-related material from Bronze Age Crete*. Saarbrücken.

Hofer, M. (2003). *The games we played: The golden age of board & table games*. Princeton Architectural Press.

Hofstadter, D. R. (1999). *Gödel, Escher, Bach: An eternal golden braid*. Basic Books. (Original work published 1979)

Huizinga, J. (2002). *Homo ludens: A study of the play-element in culture*. Routledge. (Original work published 1949)

Juul, J. (2011). *Half-real: Video games between real rules and fictional worlds*. MIT Press.

Murray, H. J. R. (1913). *A history of chess*. Oxford University Press.

Murray, H. J. R. (1952). *A history of board games other than chess*. Oxford University Press.

Parlett, D. (2018). *Oxford history of board games*. Oxford University Press.

Pritchard, D. B. (1994). *The encyclopedia of chess variations*. Games & Puzzles Publications.

Ratrout, S. (2019). A guide to checkers families and rules. Retrieved September 15, 2019, from https://www.academia.edu/28503616/A_Guide_to_Checkers_Families_and_Rules

Roberts, J. M., Arth, M. J., & Bush, R. R. (1959). Games in culture. *American Anthropologist*, *61*(4), 597–605.

Rogersdotter, E. (2011). *Gaming in Mohenjo-Daro: An archaeology of unities* (GOTARC Series B. Gothenburg Archaeological Theses No. 55) [PhD thesis, University of Gothenburg.]

Rogersdotter, E. (2015). What's left of games are boards alone: on form, incidence, and variability of engraved game boards at Vijayanagara (c. AD 1350–1565). *Heritage: Journal of Multidisciplinary Studies in Archaeology*, *3*, 457–496.

Salen, K., & Zimmerman, E. (2004). *Rules of play: Game design fundamentals*. MIT Press.

Schädler, U. (Ed.). (2007). *Jeux de l'Humanité: 5000 ans D'histoire Culturelle des Jeux de Société (Humanity games: 5000 years of the cultural history of board games)*. Slatkine.

Searle, J. R. (1980). Minds, brains, and programs. *Behavioral and Brain Sciences*, *3*(3), 417–457.

Swiny, S. (1980). *Bronze Age gaming stones from Cyprus* (pp. 54–78). Department of Antiquities, Nicosia, Cyprus. https://books.google.ca/books/about/Bronze_Age_Gaming_Stones_from_Cyprus.html?id=oPG8GwAACAAJ&redir_esc=y

Tait, W. J. (1982). *Game-boxes and accessories from the tomb of Tut'ankhamun*. Griffith Institute.

Thompson, J. M. (2000). Defining the abstract. *The Games Journal*. Retrieved August 25, 2018, from www.thegamesjournal.com/articles/DefiningtheAbstract.shtml

Townshend, P. (1980). Games of strategy: A new look at correlates and cross-cultural methods. In H. Schwartzman (Ed.), *Play and culture: 1978 Proceedings of the association for the anthropological study of play* (pp. 217–225). Leisure Press. https://anthrosource.onlinelibrary.wiley.com/doi/abs/10.1525/aa.1982.84.3.02a00190

Townshend, P. (1986). *Games in culture: A contextual analysis of the Swahili board game and its relevance to variation in African Mankala* [PhD dissertation, University of Cambridge.] https://www.repository.cam.ac.uk/handle/1810/250894

Voorhies, B. (Ed.). (2017). *Prehistoric games of North American Indians: Subarctic to Mesoamerica*. University of Utah Press.

Walker, R. A. (1990). *Sculptured mancala game boards of Sub-Saharan Africa*. [unpublished PhD dissertation, Indiana University.]
Whitehill, B. (1999). American games: A historical perspective. *Board Game Studies Journal*, *2*, 116–141.
Yalom, M. (2005). *Birth of the chess queen: A history*. HarperCollins.

Author Biographies

Sandra Danilovic is an assistant professor of game design and development at Wilfrid Laurier University. She is a computer game design theorist, qualitative researcher, and multimedia artist. Her latest arts-based research examines game design and development as a therapeutic practice for game developers. She has published on game design for mental health, machinima, and disability and virtual worlds.

Alex de Voogt is an associate professor at Drew University and cofounder of the *Board Game Studies Journal* as well as the Board Game Studies Colloquium series, which have been in existence since 1998 and 1995, respectively. He has published extensively on the history and archaeology of board games both in academic articles and books. His most recent, coauthored, book, *Ancient Egyptians at Play*, was published by Bloomsbury Academic Press in 2016.